Implementation of Benders Decomposition Method

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Large Scale Optimization

Assignment 3

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Abstract

The main objective of this work is the implementation of Benders decomposition for solving large-scale mixed-integer linear programming problems. As a case study, the method is applied to a network design problem, where the goal is to determine the optimal set of links to be added to an existing network.

The problem involves binary investment decisions and continuous flow variables, leading to a large mixed-integer formulation. Benders decomposition is used to separate these two components, resulting in a master problem that handles design decisions and a subproblem that computes optimal flows.

The subproblem is formulated as a multi-commodity network flow problem, while the master problem iteratively incorporates optimality cuts derived from dual variables. The algorithm is implemented in AMPL and tested using different initializations of the decision variables.

Finally, the results obtained through Benders decomposition are compared with those from the direct solution of the full problem, validating the correctness and effectiveness of the implementation.

1 Introduction

Benders decomposition is a classical technique for solving large-scale optimization problems with a particular block structure, typically involving both continuous and discrete decision variables [1]. The method decomposes the original problem into a master problem and one or more subproblems, allowing for a more efficient solution procedure.

The objective of this assignment is to implement the Benders decomposition algorithm and apply it to a particular problem. In this case it is a network design problem, which naturally exhibits the required structure: binary variables representing design decisions and continuous variables representing flow.

The selected application consists of determining which candidate links should be added to an existing data communications network. Once the network configuration is fixed, flows must be routed from a set of origins to a set of destinations, generating exploitation costs. The addition of new links incurs investment costs, creating a trade-off between exploitation and investment.

This setup is exactly one of the cases where Benders Decomposition method is very valuable, because it allows us to separate combinatorial decisions from continuous optimization. That is what reduces computational complexity.

The implementation is carried out in AMPL, and the results are analysed by comparing the performance of the decomposition approach with the direct solution of the full problem.

2 Algorithm Formulation

2.1 Network Design Problem as a Case Study

To implement Benders decomposition, we consider a network design problem as a test case. Let N be the set of nodes, $O \subseteq N$ the set of origins, A the set of existing arcs, and $\hat{A}$ the set of candidate arcs.

Each origin $\ell \in O$ generates a fixed amount of flow that must be distributed among a set of destination nodes. Each student was given a different configuration of candidate arcs out of the same general structure. In my case it was describes in file `benders-grafs-Page_11.jpg`:

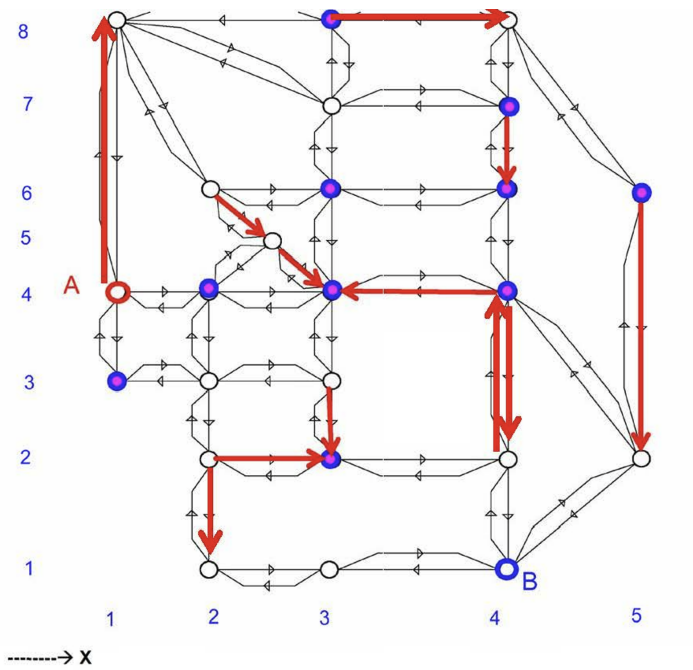

Figure 1: Configuration of candidate arcs (red arrows), origin of flows (A and B) and destination nodes (blue dots with pink-coloured centres)

Let x_{ij}^{ℓ} denote the flow through arc (i, j) associated with origin ℓ, and let $y_{ij} \in \{0, 1\}$ indicate whether a candidate arc is included in the network.

The optimization problem is defined as:

$$\min_{x,y} \sum_{\ell \in O} \sum_{(i,j) \in A \cup \hat{A}} c_{ij}^{\ell} x_{ij}^{\ell} + \sum_{(i,j) \in \hat{A}} f_{ij} y_{ij} \tag{1}$$

subject to:

$$\sum_{j:(i,j)} x_{ij}^{\ell} - \sum_{j:(j,i)} x_{ji}^{\ell} = t_i^{\ell} \quad \forall i \in N, \ \forall \ell \in O \tag{2}$$

$$x_{ij}^{\ell} \leq \rho \, y_{ij} \quad \forall (i, j) \in \hat{A}, \ \forall \ell \in O \tag{3}$$

$$x_{ij}^{\ell} \geq 0, \quad y_{ij} \in \{0, 1\} \tag{4}$$

This formulation exhibits the structure required for Benders decomposition: binary design variables y and continuous flow variables x.

In this specific instance, both investment and exploitation costs are defined as functions of the spatial coordinates of the nodes. Let (x_i, y_i) denote the coordinates of node $i \in N$. The exploitation cost associated with arc (i, j) for any origin $\ell \in O$ is given by:

$$c_{ij}^{\ell} = 100 + (x_i - x_j)^2 + (2y_i - y_j)^2 \tag{5}$$

This cost captures both a fixed component and a distance-based component, reflecting the operational cost of sending flow through the network.

The investment cost for activating a candidate arc $(i, j) \in \hat{A}$ is defined as:

$$f_{ij} = 10\,(|x_i - x_j| + |y_i - y_j|) \tag{6}$$

which corresponds to a Manhattan-distance-based cost, representing the infrastructure cost of adding a new connection.

2.2 Benders Decomposition Framework

The structure of the network design problem allows the application of Benders decomposition by separating the binary design variables y from the continuous flow variables x.

Following the formulation provided in the assignment, the constraints can be written in compact matrix form. Let B denote the node-arc incidence matrix of the network, and let x^ℓ be the vector of flows associated with origin $\ell \in O$. The flow conservation constraints can be expressed as:

$$Bx^\ell = t^\ell, \quad \forall \ell \in O \tag{7}$$

Additionally, the capacity constraints on candidate arcs can be written as:

$$x_a^\ell \leq \rho\, y_a, \quad \forall a \in \hat{A},\ \forall \ell \in O \tag{8}$$

To express both sets of constraints in a unified way, the formulation is extended by introducing slack variables and rewriting the system as:

$$\begin{pmatrix} B & 0 \\ I & I \end{pmatrix} \begin{pmatrix} x^\ell \\ \sigma^\ell \end{pmatrix} - \rho \begin{pmatrix} 0 \\ I \end{pmatrix} y = \begin{pmatrix} t^\ell \\ 0 \end{pmatrix} \tag{9}$$

where σ^ℓ are slack variables associated with the capacity constraints. This extended system allows both flow conservation and capacity constraints to be handled in a single matrix equation:

$$Bx^\ell - Fy = T^\ell \tag{10}$$

where F contains the flows and design variables, and T^ℓ is the extended right-hand side.

The dual variables associated with this system are denoted by $\theta^\ell = (u^\ell, \tau^\ell)$, where:

- u_i^ℓ are the dual variables associated with the flow conservation constraints,
- τ_a^ℓ are the dual variables associated with the capacity constraints.

Fixing the design variables to a given value $\bar{y}$, the subproblem can be written as:

$$\phi(\bar{y}) = \min_x \sum_{\ell \in O} c^{\ell\top} x^\ell \tag{11}$$

subject to:

$$Bx^\ell = t^\ell, \quad x_a^\ell \leq \rho\, \bar{y}_a, \quad x^\ell \geq 0 \tag{12}$$

Solving this problem yields both the optimal flows and the corresponding dual variables u^ℓ.

Using duality theory, the contribution of the subproblem can be expressed as:

$$c^{\ell\top} x^\ell = t^{\ell\top} u^\ell \tag{13}$$

which allows the construction of Benders optimality cuts. The master problem is then formulated as:

$$\min_{y,z} \quad z \tag{14}$$

subject to:

$$z \geq \sum_{a\in\hat{A}} f_a y_a + \sum_{\ell\in O} \left(T^\ell + Fy\right)^\top \hat{\theta}^\ell \tag{15}$$

Expanding this expression and using the structure of the dual variables, the cuts can be written as:

$$z \geq \sum_{a\in\hat{A}} f_a y_a + \sum_{\ell\in O} t^{\ell\top} u^\ell - \rho \sum_{\ell\in O} \sum_{a\in\hat{A}} \tau_a^{\ell-} y_a \tag{16}$$

where the coefficients $\tau_a^{\ell-}$ are computed as:

$$\tau_a^{\ell-} = \max\left\{0,\ u_i^\ell - u_j^\ell - c_{ij}^\ell\right\} \tag{17}$$

for each candidate arc $a = (i, j)$.

This formulation directly corresponds to the implementation, where:

- u^ℓ are obtained from the dual values of the flow conservation constraints,
- $\tau_a^{\ell-}$ are computed using the node potentials,
- each iteration adds a new Benders cut to the master problem.

The algorithm proceeds iteratively by solving the subproblem, generating cuts, and updating the master problem until convergence.

3 Results of the Implementations

The implementation of the proposed optimization problem has been carried out in two different ways: a direct resolution of the full mixed-integer problem and an implementation based on Benders decomposition. The corresponding models and execution

scripts are provided in the files `network_design.dat`, `network_design_benders.mod`, `run_original.run`, `run_benders_one.run` and `run_benders_zero.run`.

The direct solution of the problem, obtained by solving the full mixed-integer formulation using AMPL and CPLEX, yields an optimal objective value of **826640**. This value is decomposed into an **investment cost of 80 and an exploitation cost of 826560**. The optimal design consists of activating the candidate arcs $(\mathbf{4}, \mathbf{5})$, $(\mathbf{6}, \mathbf{14})$, $(\mathbf{11}, \mathbf{22})$ and $(\mathbf{14}, \mathbf{13})$, while the rest of candidate arcs remain inactive. The resulting flow configuration satisfies all demand constraints and distributes the flow efficiently through the network.

The Benders decomposition approach was implemented following the classical master-subproblem structure. Two different initializations were considered in order to analyse the behaviour of the algorithm: starting with all candidate arcs activated (`run_benders_one.run`) and starting with no candidate arcs activated (`run_benders_zero.run`).

In the case where the algorithm is initialized with $y = 1$, the method converges after 18 iterations. The evolution of the master and subproblem values shows a progressive tightening of the lower bound provided by the master problem. Initially, the gap between the subproblem evaluation and the master approximation is significant, but it decreases as new Benders cuts are added. At the final iteration, the gap becomes zero, indicating convergence to the optimal solution. The final solution matches exactly the one obtained by the direct method, both in terms of objective value (826640) and selected arcs.

Table 1: Benders decomposition iterations starting from $y_a = 1$, $\forall a \in \hat{A}$.

Iter.	Master $\bar{z}$	Subproblem z_D	Investment	Exploitation	Gap
1	826560	826770	210	826560	210
2	826570	948840	0	948840	122270
3	826570	907810	10	907800	81240
4	826580	926770	10	926760	100190
5	826580	885140	20	885120	58560
6	826590	923540	20	923520	96950
7	826590	907230	30	907200	80640
8	826600	901470	30	901440	74870
9	826600	885160	40	885120	58560
10	826610	891760	40	891720	65150
11	826610	850130	50	850080	23520
12	826610	869690	50	869640	43080
13	826620	891770	50	891720	65150
14	826620	828060	60	828000	1440
15	826630	866460	60	866400	39830
16	826630	850150	70	850080	23520
17	826640	844390	70	844320	17750
18	826640	826640	80	826560	0

Similarly, when starting from the initial solution $y = 0$, the algorithm converges after 17 iterations. In this case, the first iteration shows a very large gap, since the initial solution corresponds to a network with no additional arcs and therefore high exploitation costs. As the algorithm progresses, the master problem once again adds

more information from the subproblem through Benders cuts, eventually guiding the solution towards the optimal design. Again, convergence is achieved when the gap reaches zero, and the final solution is the same as the one obtained in the previous case and with the direct approach.

Table 2: Benders decomposition iterations starting from $y_a = 0$, $\forall a \in \hat{A}$.

Iter.	Master $\bar{z}$	Subproblem z_D	Investment	Exploitation	Gap
1	0	948840	0	948840	948840
2	826570	826680	120	826560	110
3	826570	948850	10	948840	122280
4	826570	926770	10	926760	100200
5	826580	907210	10	907200	80630
6	826580	885740	20	885720	59160
7	826590	923540	20	923520	96950
8	826590	907230	30	907200	80640
9	826600	901470	30	901440	74870
10	826600	885160	40	885120	58560
11	826610	891760	40	891720	65150
12	826610	850130	50	850080	23520
13	826620	869690	50	869640	43070
14	826620	866460	60	866400	39840
15	826650	828060	60	828000	1410
16	826640	849210	90	849120	22570
17	826640	826640	80	826560	0

A positive remark is that both initializations lead to the same optimal solution, which suggests that the implementation of the algorithm is correct and performs properly. However, the trajectory followed by the algorithm differs depending on the initial point. Starting from $y = 1$ provides a better initial feasible solution in terms of exploitation cost, while starting from $y = 0$ results in a poorer initial solution but allows the algorithm to progressively build the network structure. Despite these differences, both approaches converge in a similar number of iterations.

For the optimal network configuration, the flow values on the arcs are reported below for each origin. Only arcs with non-zero flow are shown.

For origin $\ell = 3$, the main flows are:

$$
\begin{aligned}
x^{(3)}_{1,2} &= 120, \quad x^{(3)}_{2,3} = 120, \quad x^{(3)}_{6,3} = 840, \\
x^{(3)}_{7,3} &= 240, \quad x^{(3)}_{14,6} = 720, \quad x^{(3)}_{13,14} = 360, \\
x^{(3)}_{19,7} &= 240, \quad x^{(3)}_{21,18} = 120, \quad x^{(3)}_{23,24} = 120
\end{aligned}
$$

For origin $\ell = 11$, the flows are:

$$
\begin{aligned}
x_{5,4}^{(11)} &= 120, \quad x_{8,11}^{(11)} = 240, \quad x_{12,11}^{(11)} = 720, \\
x_{13,12}^{(11)} &= 600, \quad x_{14,13}^{(11)} = 360, \quad x_{22,11}^{(11)} = 240, \\
x_{21,20}^{(11)} &= 120, \quad x_{23,22}^{(11)} = 120
\end{aligned}
$$

These flows satisfy all demand requirements and are consistent with the selected network configuration.

In terms of computational behaviour, the Benders decomposition successfully separates the combinatorial aspect of the problem from the continuous flow optimization. The subproblem is always solved efficiently as a linear network flow problem, while the master problem gradually increases in complexity as cuts are added. The convergence is achieved when no violated cuts can be generated, which corresponds to the condition that the master problem provides an exact representation of the original problem.

4 Conclusions

The objective of this project was the implementation and analysis of Benders decomposition, using a network design problem as a case study. The results obtained confirm that the decomposition has been correctly implemented, as it reproduces exactly the optimal solution of the full mixed-integer formulation and provides the same set of new arcs to add to the system. Moreover, the algorithm provides a stable convergence behaviour under different initializations, reaching the same solution from both $y = 0$ and $y = 1$ starting points. From a practical point of view, Benders decomposition is useful here because it splits the problem into two parts: deciding which links to add, and then computing the flows once the network is fixed. This not only simplifies the subproblem, which becomes a tractable network flow problem, but also provides insight into the role of dual variables in guiding the design decisions. Although the number of iterations required is moderate, the approach scales better conceptually to larger instances, where solving the full problem directly may become computationally challenging.

References

[1] Wikipedia contributors. *Benders decomposition — Wikipedia, The Free Encyclopedia*. [Online; accessed 26-April-2026]. 2026. URL: `https://en.wikipedia.org/w/index.php?title=Benders_decomposition&oldid=1349678423`.